

1. Questions

Study the following data carefully and answer the questions:

The data given below is related to the distances covered by A and B from 12 PM to 5 PM on a particular day.

Table given below shows the following data as cumulative:

Time period	The distance covered by A and B together (in km)	The distance covered by B as % of that covered by A
Till 1 PM	148	85%
Till 2 PM	266	90%
Till 3 PM	441	96%
Till 4 PM	576	92%
Till 5 PM	704	83(1/3)%

The distance covered by A from 2:00 PM to 2:30 PM is 45 km and his speed from 2:30 PM to 3:00 PM is 'M' km/h. The distance covered by B from 2:00 PM to 2:30 PM is 30 km and his speed from 2:30 PM to 3:00 PM is 'N' km/h. Find the ratio of M to N?

- 2: 3
- 25: 54
- 5: 9
- 13: 18
- None of these

2. Questions

If the ratio of the distance covered by A from 1 PM to 2PM and the distance covered by A from 4 PM to 5 PM is $(x + 2) : (y - 2)$, then find that which of the following statements is/are true?

I: The ratio of 'x' and 'y' is 1: 3.

II: The value of $(y^2 - x^3)$ is a prime number.

III: The value of $(x + y)^2$ is an odd number.

- Only I and III
- Only II and III
- Only I
- Only II
- None is true

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3. Questions

If the digits of the distance covered by A from 3 PM to 4 PM were reversed and the digits of the distance covered by B from 1 PM to 2 PM were reversed, then what would have been the ratio of the total distance covered by A to that covered by B from 12 PM to 5 PM?

- a. 11: 7
- b. 122: 107
- c. 61: 47
- d. None of these
- e. 366: 347

4. Questions

The distance covered by B from 3 PM to 4 PM is 'X'% of the distance covered by A from 3 PM to 4 PM and the difference between the distances covered by A and B from 12 PM to 1 PM is 'Y' km. If the distance covered by A from 5 PM to 6 PM is 'X' km and the distance covered by B from 5 PM to 6 PM is '7Y' km, then find the average of distances covered by A and B from 5 PM to 6 PM?

- a. 78 km
- b. 86 km
- c. 85 km
- d. 82 km
- e. 79 km

5. Questions

If the distance covered by C from 12 PM to 1 PM is ____ km and the distance covered by C from 1 PM to 2 PM is ____ km, then the distance covered by C till 2 PM will be more than 70% of the distance covered by A till 2 PM.

Find that which of the following can be filled in the blanks in any order?

P: 60, 45

Q: 33, 67

R: 45, 53

- a. Only P and Q
- b. Only Q and R
- c. Only Q
- d. Only P
- e. Only P and R

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6. Questions

Study the following data carefully and answer the questions:

There are three bags A, B and C and each bag contains 5 rupee coins and 10 rupee coins.

In bag A, the number of 5 rupee coins is 3 more than the number of 10 rupee coins and the probability of taking out two 5 rupee coins is $\frac{14}{39}$. The ratio of the number of 5 rupee coins in bag A to that in bag B is 2: 1 and the total value in bag B is Rs. 80. The ratio of the number of 10 rupee coins in bag B to that in bag C is 2: 1 and the probability of taking out one 5 rupee coin from bag C is $\frac{2}{3}$.

If 4 coins are taken out from bag B, then what is the probability that at-least two of them are 5 rupee coins?

- a. $\frac{19}{42}$
- b. $\frac{23}{42}$
- c. $\frac{17}{42}$
- d. $\frac{25}{42}$
- e. None of these

7. Questions

When M number of 5 rupee coins and (M + 1) number of 10 rupee coins of bag A are put into bag C, the total value in bag C becomes Rs 100. Find the probability of taking out two 10 rupee coins from bag A after the given process?

- a. $\frac{1}{14}$
- b. $\frac{1}{7}$
- c. $\frac{1}{56}$
- d. $\frac{1}{28}$
- e. $\frac{1}{35}$

8. Questions

Which of the following statements is/are not true?

I: The ratio of total amount in bag A to that in bag C is 3: 2.

II: The probability of taking out one 5 rupee coin from bag B is $\frac{4}{10}$.

III: The total amount in all the three bags together is more than Rs. 250.

- a. Only II and III
- b. None is true
- c. Only III
- d. Only I and II

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e. Only II

9. Questions

The ratio of the number of 5 rupee coins in bag B to that in bag C is $P:Q$. If the number of 5 rupee coins in bag D is $(P + Q)$ and the number of 10 rupee coins in bag D is $(Q^2 - P^2)$, then find the probability of taking out two 10 rupee coins from bag D (in % and up to two decimal places)?

- a. 22.22%
- b. 33.33%
- c. 16.67%
- d. 13.33%
- e. None of these

10. Questions

When X number of 10 rupee coins is put into bag C, the probability of taking out one 5 rupee coin from bag C becomes $6/11$. If the given process were not done and X number of 5 rupee coins were taken out from bag C, then what would have been the probability of taking out one 5 rupee coin from bag C (after taking out X number of 5 rupee coins)?

- a. $2/5$
- b. $4/7$
- c. $3/5$
- d. Can't be determined
- e. $2/7$

11. Questions

The radius of cylinder A is 7 cm more than that of cylinder B and the ratio of the curved surface area of cylinder A to that of cylinder B is 1: 2. If the height of cylinder A is ___ % of that of cylinder B, then the radius of cylinder B will be ___ cm.

Find that which of the following values can't be filled in the blanks? (in the same order)

P: $33(1/3)$, 10.5

Q: 25, 7

R: 40, 21

- a. Only Q
- b. Only P and R
- c. Only R
- d. Only Q and R

e. Only P

12. Questions

A and B together can complete a work in 12 hours, while A and C together can complete the same work in $12\frac{6}{7}$ hours. If the ratio of B's efficiency to C's efficiency is 5: 4, then find that which of following statements is/are true?

P: A alone can complete 55% part of work in 9 hours 9 minutes.

Q: B and C together can complete the work in 20 hours.

R: Ratio of A's efficiency to C's efficiency is 2: 5.

- a. Only Q
- b. Only P and R
- c. Only R
- d. Only Q and R
- e. Only P

13. Questions

The difference between the speeds of trains A and B is 25.2 km/h. The lengths of trains A and B are 115 m and 'd' m respectively. Train A crosses train B in 45 seconds while running in same direction. Which of the following can't be determined?

P: In what time, train A will cross (d + 60) m long tunnel?

Q: In what time, train B will cross (d + 50) m long platform?

R: In what time, trains A and B will cross each other while running in opposite directions?

- a. Only R
- b. Only Q and R
- c. None can be determined
- d. Only P
- e. Only P and Q

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14. Questions

A shopkeeper marked up an article by 25%. The profit amount is $16\frac{2}{3}\%$ of its selling price. The discount given is M% and the profit per cent on the cost price is N%. If he offered new discount per cent were (M + N)%, then what would have been the profit/loss per cent on the cost price of the article?

- a. 7.5%
- b. 12.5%

- c. 5%
- d. 10%
- e. Cannot be determined

15. Questions

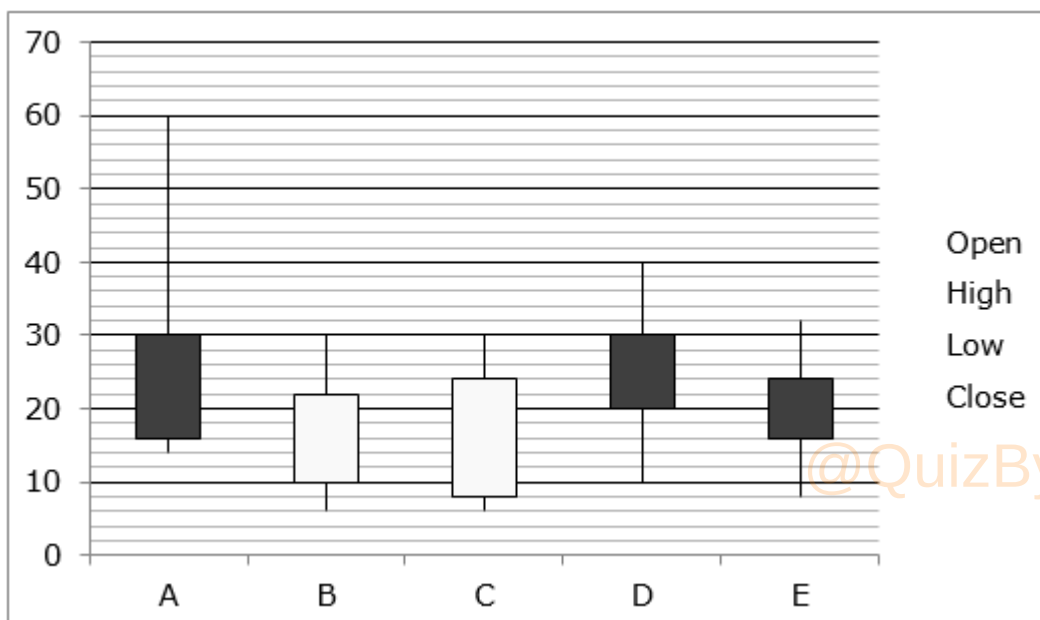
Two boats A and B are sailing in a river, the speeds of boats A and B in still water are 'a' m/s and 'b' m/s respectively, where ($a < b$) and the speed of stream is 1 m/s less than the difference between the speeds of boats A and B in still water. If the ratio of the downstream speed of boat A to the upstream speed of boat B is 9: 8, then find the relation between 'a' and 'b'.

- a. $9b - 8a = 51$
- b. $9a - 8b = 17$
- c. $8b - 9a = 17$
- d. $8a - 9b = 51$
- e. None of these

16. Questions

Study the following data carefully and answer the questions:

The OHLC (Open-High-Low-Close) chart given below is related to the quantities (in kg) of oranges and apples sold by a shopkeeper in five days Monday, Tuesday, Wednesday, Thursday and Friday.



Notes:

- 1: (High value + Low value) in a candle = (Total quantity of oranges and apples sold together)
- 2: (Close value – Open value) in a candle = The difference between Quantity of sold oranges and the Quantity of sold apples
- 3: White candle: - The quantity of sold oranges is more than the quantity of sold apples

4: Black candle: - The quantity of sold oranges is less than the quantity of sold apples

If the ratio of the quantity of fresh oranges to the quantity of stale oranges sold on Wednesday is 8: 5 and the ratio of the quantity of fresh apples to the quantity of stale apples sold on Wednesday is 3: 2, then find the total quantity of fresh fruits sold on Wednesday is what per cent more than the total quantity of stale fruits sold on Wednesday?

- a. 44.17%
- b. 57.14%
- c. 51.47%
- d. Can't be determined
- e. None of these

17. Questions

If the price of oranges on Tuesday was Rs. 45 per kg, the price of apples on Tuesday was Rs. 50 per kg, the price of oranges on Thursday was Rs. 50 per kg and the price of apples on Thursday was Rs. 40 per kg, then find that which of the following statements is/are true?

I: The ratio of the amount received on Tuesday by selling apples to that received on Thursday by selling apples is 1: 2.

II: The total amount received by selling oranges on Tuesday and Thursday is Rs. 2080.

III: The amount received by selling oranges on Tuesday is 90% of that received by selling apples on Thursday.

- a. Only I and III
- b. Only II and III
- c. Only I
- d. Only II
- e. All are true

18. Questions

The quantity of apples sold on Friday is M% more than the quantity of oranges sold on Friday and the quantity of oranges sold on Wednesday is N% more than the quantity of apples sold on Wednesday. Find that which of the following relations between M and N is correct?

I: $M = (N/2) - 20$

II: $3M - N = 10$

III: $N/10 = (M/10) + 10$

- a. Only I and II
- b. Only III

- c. Only II and III
- d. None is correct
- e. Only I

19. Questions

If the price of oranges on Monday is Rs. X per kg, the price of apples on Monday is Rs. (X + Y) per kg and the ratio of the total amount received by selling oranges on Monday to that received by selling apples on Monday is 6: 11, then find the ratio of X to Y?

- a. Can't be determined
- b. 4: 3
- c. 5: 2
- d. 3: 2
- e. 4: 1

20. Questions

If the quantities of apples sold on each of the given days are described in a pie chart, then find the difference between the degree distribution of the quantity of apples sold on Monday and the degree distribution of the quantity of apples sold on Friday?

- a. 72°
- b. 36°
- c. 60°
- d. 102°
- e. 50°

Explanations:**1. Questions**

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Since, the ratio of the distance covered by A to that covered by B till 1 PM = 100: 85 = 20: 17

So, the distance covered by A till 1 PM = $148 * [20/(20 + 17)] = 80$ km

And the distance covered by B till 1 PM = $148 - 80 = 68$ km

Similarly, we can calculate for other time periods as well:

Time period	The distance covered by A (in km)	The distance covered by B (in km)
Till 1 PM	80	68
Till 2 PM	140	126
Till 3 PM	225	216
Till 4 PM	300	276
Till 5 PM	384	320

Then solve

Time period	The distance covered by A (in km)	The distance covered by B (in km)
From 12 PM to 1 PM	80	68
From 1 PM to 2 PM	$140 - 80 = 60$	$126 - 68 = 58$
From 2 PM to 3 PM	$225 - 140 = 85$	$216 - 126 = 90$
From 3 PM to 4 PM	$300 - 225 = 75$	$276 - 216 = 60$
From 4 PM to 5 PM	$384 - 300 = 84$	$320 - 276 = 44$

Answer: A

Since, the total distance covered by A from 2:00 PM to 3:00 PM = 85 km

And the distance covered by A from 2:00 PM to 2:30 PM = 45 km

So, the speed of A from 2:30 PM to 3:00 PM = $M = (85 - 45)/(1/2) = 80$ km/h

Since, the total distance covered by B from 2:00 PM to 3:00 PM = 90 km

And the distance covered by B from 2:00 PM to 2:30 PM = 30 km

So, the speed of B from 2:30 PM to 3:00 PM = $N = (90 - 30)/(1/2) = 120$ km/h

Required ratio = $M : N = 80 : 120 = 2 : 3$

Hence, option A is correct.

2. Questions

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Since, the ratio of the distance covered by A to that covered by B till 1 PM = $100 : 85 = 20 : 17$

So, the distance covered by A till 1 PM = $148 * [20/(20 + 17)] = 80$ km

And the distance covered by B till 1 PM = $148 - 80 = 68$ km

Similarly, we can calculate for other time periods as well:

Time period	The distance covered by A (in km)	The distance covered by B (in km)
Till 1 PM	80	68
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Till 4 PM	300	276
Till 5 PM	384	320

Then solve

Time period	The distance covered by A (in km)	The distance covered by B (in km)
From 12 PM to 1 PM	80	68
From 1 PM to 2 PM	$140 - 80 = 60$	$126 - 68 = 58$
From 2 PM to 3 PM	$225 - 140 = 85$	$216 - 126 = 90$
From 3 PM to 4 PM	$300 - 225 = 75$	$276 - 216 = 60$
From 4 PM to 5 PM	$384 - 300 = 84$	$320 - 276 = 44$

Answer: C

Since, the distance covered by A from 1 PM to 2 PM = 60 km

And the distance covered by A from 4 PM to 5 PM = 84 km

So, $(x + 2) : (y - 2) = 60 : 84$

$(x + 2) : (y - 2) = 5 : 7$

Since, $x + 2 = 5$

So, $x = 3$

Since, $y - 2 = 7$

So, $y = 9$

From I:

Ratio of 'x' to 'y' = $3 : 9 = 1 : 3$

So, I is true.

From II:

Value of $(y^2 - x^3) = (9^2 - 3^3) = (81 - 27) = 54$

Since, 54 is not a prime number.

So, II is not true.

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From III:

Value of $(x + y)^2 = (3 + 9)^2 = 144$

Since, 144 is not an odd number.

So, III is not true.

Hence, only I is true.

Hence, option C is correct.

3. Questions

Since, the ratio of the distance covered by A to that covered by B till 1 PM = 100: 85 = 20: 17

So, the distance covered by A till 1 PM = $148 * [20/(20 + 17)] = 80$ km

And the distance covered by B till 1 PM = $148 - 80 = 68$ km

Similarly, we can calculate for other time periods as well:

Time period	The distance covered by A (in km)	The distance covered by B (in km)
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From 2 PM to 3 PM	$225 - 140 = 85$	$216 - 126 = 90$
From 3 PM to 4 PM	$300 - 225 = 75$	$276 - 216 = 60$
From 4 PM to 5 PM	$384 - 300 = 84$	$320 - 276 = 44$

Answer: E

If the digits of the distance covered by A from 3 PM to 4 PM were reversed:

The distance covered by A from 3 PM to 4 PM would have been = 57 km

And the total distance covered by A from 12 PM to 5 PM would have been:

$80 + 60 + 85 + 57 + 84 = 366$ km

If the digits of the distance covered by B from 1 PM to 2 PM were reversed:

The distance covered by B from 1 PM to 2 PM would have been = 85 km

And the total distance covered by B from 12 PM to 5 PM would have been:

$$68 + 85 + 90 + 60 + 44 = 347 \text{ km}$$

Required ratio = 366: 347

Hence, option E is correct.

4. Questions

Since, the ratio of the distance covered by A to that covered by B till 1 PM = 100: 85 = 20: 17

So, the distance covered by A till 1 PM = $148 * [20/(20 + 17)] = 80 \text{ km}$

And the distance covered by B till 1 PM = $148 - 80 = 68 \text{ km}$

Similarly, we can calculate for other time periods as well:

Time period	The distance covered by A (in km)	The distance covered by B (in km)
Till 1 PM	80	68
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Then solve

Time period	The distance covered by A (in km)	The distance covered by B (in km)
From 12 PM to 1 PM	80	68
From 1 PM to 2 PM	$140 - 80 = 60$	$126 - 68 = 58$
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From 3 PM to 4 PM	$300 - 225 = 75$	$276 - 216 = 60$
From 4 PM to 5 PM	$384 - 300 = 84$	$320 - 276 = 44$

Answer: D

Since, the distance covered by A from 3 PM to 4 PM = 75 km

And the distance covered by B from 3 PM to 4 PM = 60 km

So, $X = (60/75) * 100 = 80\%$

Since, the distance covered by A from 12 PM to 1 PM = 80 km

And the distance covered by B from 12 PM to 1 PM = 68 km

So, $Y = 80 - 68 = 12$ km

Since, the distance covered by A from 5 PM to 6 PM = X km = 80 km

And the distance covered by B from 5 PM to 6 PM = $7Y = 7 * 12 = 84$ km

So, the required average = $(80 + 84)/2 = 82$ km

Hence, option D is correct.

5. Questions

Since, the ratio of the distance covered by A to that covered by B till 1 PM = 100: 85 = 20: 17

So, the distance covered by A till 1 PM = $148 * [20/(20 + 17)] = 80$ km

And the distance covered by B till 1 PM = $148 - 80 = 68$ km

Similarly, we can calculate for other time periods as well:

Time period	The distance covered by A (in km)	The distance covered by B (in km)
Till 1 PM	80	68
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Then solve

Time period	The distance covered by A (in km)	The distance covered by B (in km)
From 12 PM to 1 PM	80	68
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From 3 PM to 4 PM	$300 - 225 = 75$	$276 - 216 = 60$
From 4 PM to 5 PM	$384 - 300 = 84$	$320 - 276 = 44$

Answer: A

The distance covered by A till 2 PM = $80 + 60 = 140$ km

From P:

The distance covered by C till 2 PM = $60 + 45 = 105$ km

Since, the distance covered by C till 2 PM is $(105/140) * 100 = 75\%$ of that covered by A till 2 PM.

So, the values in P can be filled in the blanks in any order.

From Q:

The distance covered by C till 2 PM will be = $33 + 67 = 100$ km

Since, the distance covered by C till 2 PM is $(100/140) * 100 = 71.42\%$ of that covered by A till 2 PM.

So, the values in Q can be filled in the blanks in any order.

From R:

If the distance covered by C from 12 PM to 1 PM is 45 km or 53 km and that covered by him from 1 PM to 2 PM is 45 km or 53 km:

The distance covered by C till 2 PM will be = $45 + 53 = 98$ km

Since, the distance covered by C till 2 PM is $(98/140) * 100 = 70\%$ of that covered by A till 2 PM.

So, the values in R can't be filled in the blanks.

Hence, the values in P and Q can be filled in the blanks.

Hence, option A is correct.

6. Questions

Let the number of 10 rupee coins in bag A = x

So, the number of 5 rupee coins in bag B = $(x + 3)$

Since, the probability of taking out two 5 rupee coins = $14/39$

So,

$$\Rightarrow {}^{(x+3)}C_2 / {}^{(2x+3)}C_2 = 14/39$$

$$\Rightarrow [(x+3)(x+2)] / [(2x+3)(2x+2)] = 14/39$$

$$\Rightarrow (x^2 + 5x + 6) / (4x^2 + 10x + 6) = 14/39$$

$$\Rightarrow 39x^2 + 195x + 234 = 56x^2 + 140x + 84$$

$$\Rightarrow 17x^2 - 55x - 150 = 0$$

$$\Rightarrow 17x^2 - 85x + 30x - 150 = 0$$

$$\Rightarrow 17x(x-5) + 30(x-5) = 0$$

$$\Rightarrow x = 5$$

The number of 5 rupee coins in bag A = $5 + 3 = 8$

The number of 10 rupee coins in bag B = 5

Since, the ratio of the number of 5 rupee coins in bag A to that in bag B = 2: 1

So, the number of 5 rupee coins in bag B = $8 * (1/2) = 4$

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Since, the total value in bag B is Rs. 80.

So, the number of 10 rupee coins in bag B = $(80 - 4 * 5)/10 = 6$

Since, the ratio of the number of 10 rupee coins in bag B to that in bag C = 2: 1

So, the number of 10 rupee coins in bag C = $6 * (1/2) = 3$

Let the number of 5 rupee coins in bag C = y

Since, the probability of taking out one 5 rupee coin from bag C = $2/3$

So,

$$y/(y + 3) = 2/3$$

$$3y = 2y + 6$$

$$y = 6$$

The number of 5 rupee coins in bag C = 6

Bag	Number of 5 rupee coins	Number of 10 rupee coins
A	8	5
B	4	6
C	6	3

Answer: B

The number of ways to taking out at-least two 5 rupee coins from bag B:

(Two 5 rupee coins and Two 10 rupee coins) or (Three 5 rupee coins and one 10 rupee coin) or (Four 5 rupee coins and Zero 10 rupee coins)

So, the required probability = $[(4C2 * 6C2)/10C4] + [(4C3 * 6C1)]/10C4 + [(4C4 * 6C0)/10C4]$

$$= [(6 * 15)/210] + [(4 * 6)/210] + [(1 * 1)/210]$$

$$= [90/210] + [24/210] + [1/210]$$

$$= 115/210$$

$$= 23/42$$

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Hence, option B is correct.

7. Questions

Let the number of 10 rupee coins in bag A = x

So, the number of 5 rupee coins in bag B = (x + 3)

Since, the probability of taking out two 5 rupee coins = $14/39$

So,

$$\Rightarrow {}^{(X+3)}C_2 / {}^{(2X+3)}C_2 = 14/39$$

$$\Rightarrow [(x+3)(x+2)] / [(2x+3)(2x+2)] = 14/39$$

$$\Rightarrow (x^2 + 5x + 6) / (4x^2 + 10x + 6) = 14/39$$

$$\Rightarrow 39x^2 + 195x + 234 = 56x^2 + 140x + 84$$

$$\Rightarrow 17x^2 - 55x - 150 = 0$$

$$\Rightarrow 17x^2 - 85x + 30x - 150 = 0$$

$$\Rightarrow 17x(x-5) + 30(x-5) = 0$$

$$\Rightarrow x = 5$$

The number of 5 rupee coins in bag A = $5 + 3 = 8$

The number of 10 rupee coins in bag B = 5

Since, the ratio of the number of 5 rupee coins in bag A to that in bag B = 2: 1

So, the number of 5 rupee coins in bag B = $8 * (1/2) = 4$

Since, the total value in bag B is Rs. 80.

So, the number of 10 rupee coins in bag B = $(80 - 4 * 5) / 10 = 6$

Since, the ratio of the number of 10 rupee coins in bag B to that in bag C = 2: 1

So, the number of 10 rupee coins in bag C = $6 * (1/2) = 3$

Let the number of 5 rupee coins in bag C = y

Since, the probability of taking out one 5 rupee coin from bag C = $2/3$

So,

$$y / (y + 3) = 2/3$$

$$3y = 2y + 6$$

$$y = 6$$

The number of 5 rupee coins in bag C = 6

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Bag	Number of 5 rupee coins	Number of 10 rupee coins
A	8	5
B	4	6
C	6	3

Answer: D

Since, the initial number of 5 rupee coins in bag C = 6

And the initial number of 10 rupee coins in bag C = 3

So,

$$[(6 + M) * 5] + [(3 + M + 1) * 10] = 100$$

$$30 + 5M + 40 + 10M = 100$$

$$M = 2$$

Since, the initial number of 5 rupee coins in bag A = 8

So, the new number of 5 rupee coins in bag A = $8 - 2 = 6$

Since, the initial number of 10 rupee coins in bag B = 5

So, the new number of 10 rupee coins in bag B = $5 - (2 + 1) = 2$

Now, the probability of taking out two 10 rupee coins from bag A = $\frac{{}^2C_2}{(6 + 2)C_2}$

$$= \frac{{}^2C_2}{8C_2}$$

$$= \frac{1}{[(8 * 7)/2]}$$

$$= \frac{1}{28}$$

Hence, option D is correct.

8. Questions

Let the number of 10 rupee coins in bag A = x

So, the number of 5 rupee coins in bag B = (x + 3)

Since, the probability of taking out two 5 rupee coins = $\frac{14}{39}$

So,

$$\Rightarrow \frac{{}^{(x+3)}C_2}{{}^{(2x+3)}C_2} = \frac{14}{39}$$

$$\Rightarrow \frac{[(x + 3)(x + 2)]}{[(2x + 3)(2x + 2)]} = \frac{14}{39}$$

$$\Rightarrow \frac{(x^2 + 5x + 6)}{(4x^2 + 10x + 6)} = \frac{14}{39}$$

$$\Rightarrow 39x^2 + 195x + 234 = 56x^2 + 140x + 84$$

$$\Rightarrow 17x^2 - 55x - 150 = 0$$

$$\Rightarrow 17x^2 - 85x + 30x - 150 = 0$$

$$\Rightarrow 17x(x - 5) + 30(x - 5) = 0$$

$$\Rightarrow x = 5$$

The number of 5 rupee coins in bag A = $5 + 3 = 8$

The number of 10 rupee coins in bag B = 5

Since, the ratio of the number of 5 rupee coins in bag A to that in bag B = 2: 1

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So, the number of 5 rupee coins in bag B = $8 * (1/2) = 4$

Since, the total value in bag B is Rs. 80.

So, the number of 10 rupee coins in bag B = $(80 - 4 * 5)/10 = 6$

Since, the ratio of the number of 10 rupee coins in bag B to that in bag C = 2: 1

So, the number of 10 rupee coins in bag C = $6 * (1/2) = 3$

Let the number of 5 rupee coins in bag C = y

Since, the probability of taking out one 5 rupee coin from bag C = $2/3$

So,

$$y/(y + 3) = 2/3$$

$$3y = 2y + 6$$

$$y = 6$$

The number of 5 rupee coins in bag C = 6

Bag	Number of 5 rupee coins	Number of 10 rupee coins
A	8	5
B	4	6
C	6	3

Answer: C

From I:

The total **amount** in bag A = $(8 * 5) + (5 * 10) = 40 + 50 = \text{Rs. } 90$

The total **amount** in bag C = $(6 * 5) + (3 * 10) = 30 + 30 = \text{Rs. } 60$

Since, the required ratio = $90: 60 = 3: 2$

So, I is true.

From II:

The number of 5 rupee coins in bag B = 4

The number of 10 rupee coins in bag B = 6

Since, the required probability = $4/(4 + 6) = 4/10$

So, II is true.

From III:

The total value in bag A = $(8 * 5) + (5 * 10) = 40 + 50 = \text{Rs. } 90$

The total value in bag B = $(4 * 5) + (6 * 10) = 20 + 60 = \text{Rs. } 80$

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The total value in bag C = $(6 * 5) + (3 * 10) = 30 + 30 = \text{Rs. } 60$

Since, the total value in all the three bags together = $90 + 80 + 60 = \text{Rs. } 230$

So, III is not true.

Hence, only III is not true.

Hence, option C is correct.

9. Questions

Let the number of 10 rupee coins in bag A = x

So, the number of 5 rupee coins in bag B = $(x + 3)$

Since, the probability of taking out two 5 rupee coins = $14/39$

So,

$$\Rightarrow {}^{(x+3)}C_2 / {}^{(2x+3)}C_2 = 14/39$$

$$\Rightarrow [(x+3)(x+2)] / [(2x+3)(2x+2)] = 14/39$$

$$\Rightarrow (x^2 + 5x + 6) / (4x^2 + 10x + 6) = 14/39$$

$$\Rightarrow 39x^2 + 195x + 234 = 56x^2 + 140x + 84$$

$$\Rightarrow 17x^2 - 55x - 150 = 0$$

$$\Rightarrow 17x^2 - 85x + 30x - 150 = 0$$

$$\Rightarrow 17x(x-5) + 30(x-5) = 0$$

$$\Rightarrow x = 5$$

The number of 5 rupee coins in bag A = $5 + 3 = 8$

The number of 10 rupee coins in bag B = 5

Since, the ratio of the number of 5 rupee coins in bag A to that in bag B = 2: 1

So, the number of 5 rupee coins in bag B = $8 * (1/2) = 4$

Since, the total value in bag B is Rs. 80.

So, the number of 10 rupee coins in bag B = $(80 - 4 * 5) / 10 = 6$

Since, the ratio of the number of 10 rupee coins in bag B to that in bag C = 2: 1

So, the number of 10 rupee coins in bag C = $6 * (1/2) = 3$

Let the number of 5 rupee coins in bag C = y

Since, the probability of taking out one 5 rupee coin from bag C = $2/3$

So,

$$y / (y + 3) = 2/3$$

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$$3y = 2y + 6$$

$$y = 6$$

The number of 5 rupee coins in bag C = 6

Bag	Number of 5 rupee coins	Number of 10 rupee coins
A	8	5
B	4	6
C	6	3

Answer: A

The ratio of the number of 5 rupee coins in bag B to that in bag C = P: Q = 4: 6 = 2: 3

Since, the number of 5 rupee coins in bag D = P + Q = 2 + 3 = 5

And the number of 10 rupee coins in bag D = $Q^2 - P^2 = 3^2 - 2^2 = 5$

So, the required probability = $\frac{5C_2}{10C_2} = \frac{(5 * 4)}{(10 * 9)} = \frac{2}{9}$

= 22.22% (Approx. up to two decimal points)

Hence, option A is correct.

10. Questions

Let the number of 10 rupee coins in bag A = x

So, the number of 5 rupee coins in bag B = (x + 3)

Since, the probability of taking out two 5 rupee coins = 14/39

So,

$$\Rightarrow \frac{{}^{(x+3)}C_2}{{}^{(2x+3)}C_2} = \frac{14}{39}$$

$$\Rightarrow \frac{[(x+3)(x+2)]}{[(2x+3)(2x+2)]} = \frac{14}{39}$$

$$\Rightarrow \frac{(x^2 + 5x + 6)}{(4x^2 + 10x + 6)} = \frac{14}{39}$$

$$\Rightarrow 39x^2 + 195x + 234 = 56x^2 + 140x + 84$$

$$\Rightarrow 17x^2 - 55x - 150 = 0$$

$$\Rightarrow 17x^2 - 85x + 30x - 150 = 0$$

$$\Rightarrow 17x(x - 5) + 30(x - 5) = 0$$

$$\Rightarrow x = 5$$

The number of 5 rupee coins in bag A = 5 + 3 = 8

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The number of 10 rupee coins in bag B = 5

Since, the ratio of the number of 5 rupee coins in bag A to that in bag B = 2: 1

So, the number of 5 rupee coins in bag B = $8 * (1/2) = 4$

Since, the total value in bag B is Rs. 80.

So, the number of 10 rupee coins in bag B = $(80 - 4 * 5)/10 = 6$

Since, the ratio of the number of 10 rupee coins in bag B to that in bag C = 2: 1

So, the number of 10 rupee coins in bag C = $6 * (1/2) = 3$

Let the number of 5 rupee coins in bag C = y

Since, the probability of taking out one 5 rupee coin from bag C = $2/3$

So,

$$y/(y + 3) = 2/3$$

$$3y = 2y + 6$$

$$y = 6$$

The number of 5 rupee coins in bag C = 6

Bag	Number of 5 rupee coins	Number of 10 rupee coins
A	8	5
B	4	6
C	6	3

Answer: B

Since, the initial number of 5 rupee coins in bag C = 6

And the initial number of 10 rupee coins in bag C = 3

So,

$$6/(6 + 3 + X) = 6/11$$

$$11 = 9 + X$$

$$X = 2$$

If the given process were not done and X number of 5 rupee coins were taken out from bag C:

The new number of 5 rupee coins in bag C would have been = $6 - X = 6 - 2 = 4$

Since, the number of 10 rupee coins in bag C = 3

So, the required probability of would have been = $4/(4 + 3) = 4/7$

Hence, option B is correct.

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11. Questions

Answer: B

Let the radius of cylinder B = 'x' cm

So, the radius of cylinder A = (x + 7) cm

The ratio of the curved surface area of cylinder A to that of cylinder B = 1: 2

From P:

The ratio of the height of cylinder A to that of cylinder B = $33\frac{1}{3}$: 100 = 1: 3

Let the heights of cylinders A and B be 'y' cm and '3y' cm respectively.

So,

$$[2\pi * (x + 7) * y] / [2\pi * x * 3y] = 1/2$$

$$(x + 7)/(3x) = 1/2$$

$$2x + 14 = 3x$$

$$x = 14$$

So, the values in P can't be filled in the blanks.

From Q:

The ratio of the height of cylinder A to that of cylinder B = 25: 100 = 1: 4

Let the heights of cylinders A and B be 'y' cm and '4y' cm respectively.

So,

$$[2\pi * (x + 7) * y] / [2\pi * x * 4y] = 1/2$$

$$(x + 7)/(4x) = 1/2$$

$$2x + 14 = 4x$$

$$x = 7$$

So, the values in Q can be filled in the blanks.

From R:

The ratio of the height of cylinder A to that of cylinder B = 40: 100 = 2: 5

Let the heights of cylinders A and B be '2y' cm and '5y' cm respectively.

So,

$$[2\pi * (x + 7) * 2y] / [2\pi * x * 5y] = 1/2$$

$$(2x + 14)/(5x) = 1/2$$

$$4x + 28 = 5x$$

$$x = 28$$

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So, the values in R can't be filled in the blanks.

Hence, the values in P and R can't be filled in the blanks.

Hence, option B is correct.

12. Questions

Answer: A

Since, the ratio of B's efficiency to C's efficiency = 5: 4

So, the ratio of time, in which B alone and C alone can complete the work will be 4: 5 respectively.

Let the time, in which B alone and C alone can complete the work be '4t' hours and '5t' hours respectively.

Since, A and B together can complete the work in 12 hours.

$$\text{So, } (1/A) + (1/4t) = (1/12) \text{ -----(1)}$$

Since, A and C together can complete the work in $12\frac{6}{7}$ hours.

$$\text{So, } (1/A) + (1/5t) = (7/90) \text{ -----(2)}$$

From equations (1) and (2):

$$(1/12) - (1/4t) = (7/90) - (1/5t)$$

$$(1/12) - (7/90) = (1/4t) - (1/5t)$$

$$(15 - 14)/180 = (5 - 4)/20t$$

$$t = 9$$

From equation (1):

$$(1/A) + (1/36) = (1/12)$$

$$1/A = (1/12) - (1/36)$$

$$1/A = (3 - 1)/36$$

$$A = 18$$

The time, in which A alone can complete the work = 18 hours

The time, in which B alone can complete the work = $4 * 9 = 36$ hours

The time, in which C alone can complete the work = $5 * 9 = 45$ hours

From P:

Since, the time, in which A alone can complete the work = 18 hours

So, the time, in which A alone can complete 55% part of work = 55% of 18 = 9.9 hours = 9 hours 54 minutes

So, P is not true.

From Q:

Since, the part of work completed by B and C together in 1 hour = $(1/36) + (1/45) = (5 + 4)/180 = 1/20$

So, the time, in which B and C together can complete the work = 20 hours

So, Q is true.

From R:

Since, the ratio of time, in which A alone and C alone can complete the work = 18: 45 = 2: 5

So, the ratio of A's efficiency to C's efficiency = 5: 2

So, R is not true.

Hence, only Q is true.

Hence, option A is correct.

13. Questions

Answer: C

Since, train A crosses train B in 45 seconds.

So, the speed of train A will be more than the speed of train B.

Let the speed of train B = 'x' m/s

So, the speed of train A = $x + 25.2 * (5/18) = (x + 7)$ m/s

From the question:

$$(x + 7) - x = (115 + d)/45$$

$$7 = (115 + d)/45$$

$$d = 200$$

From P:

Since, the speed of train A can't be calculated.

So, the required time can't be determined.

From Q:

Since, the speed of train B can't be calculated.

So, the required time can't be determined.

From R:

Since, the speeds of trains A and B can't be calculated.

So, the required time can't be determined

Hence, none can be determined.

Hence, option C is correct.

14. Questions

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Answer: C

Let the CP of the article = Rs. $100x$

So, the MP of the article = 125% of $100x$ = Rs. $125x$

Also let the SP of the article = Rs. y

Since, the profit is $16\frac{2}{3}\%$ of the SP.

So,

$$(y - 100x) = 16\frac{2}{3}\% \text{ of } y$$

$$y - 100x = y/6$$

$$y - y/6 = 100x$$

$$5y/6 = 100x$$

$$y = 120x$$

The CP of the article = Rs. $100x$

The MP of the article = Rs. $125x$

The SP of the article = Rs. $120x$

$$\text{The discount given} = M = [(125x - 120x)/125x] * 100 = 4\%$$

$$\text{The profit per cent on the cost price} = N = [(120x - 100x)/100x] * 100 = 20\%$$

If the discount per cent were $(M + N)\%$.

$$\text{New discount per cent would have been} = 4 + 20 = 24\%$$

$$\text{The new SP would have been} = (100 - 24)\% \text{ of } 125x = \text{Rs. } 95x$$

$$\text{And the loss per cent on the cost price would have been} = [(100x - 95x)/100x] * 100 = 5\%$$

Hence, option C is correct.

15. Questions

Answer: C

Since, the speed of boat A in still water = 'a' m/s

And the speed of boat B in still water = 'b' m/s

$$\text{So, the speed of stream} = (b - a - 1) \text{ m/s}$$

$$\text{The downstream speed of boat A} = (a + b - a - 1) = (b - 1) \text{ m/s}$$

$$\text{And the upstream speed of boat B} = (b - b + a + 1) = (a + 1) \text{ m/s}$$

Since, the ratio of the downstream speed of boat A to the upstream speed of boat B = 9: 8

So,

$$\Rightarrow (b - 1)/(a + 1) = 9/8$$

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$$\Rightarrow 8b - 8 = 9a + 9$$

$$\Rightarrow 8b - 9a = 17$$

Hence, option C is correct.

16. Questions

Days	Total quantity of oranges and apples sold together (in kg)	The difference between Quantity of sold oranges and the Quantity of sold apples
Monday	$60 + 14 = 74$	$30 - 16 = 14$
Tuesday	$30 + 6 = 36$	$22 - 10 = 12$
Wednesday	$30 + 6 = 36$	$24 - 8 = 16$
Thursday	$40 + 10 = 50$	$30 - 20 = 10$
Friday	$32 + 8 = 40$	$24 - 16 = 8$

Days	Quantity of sold oranges (in kg)	Quantity of sold apples (in kg)
Monday	$(74 - 14)/2 = 30$	$(74 + 14)/2 = 44$
Tuesday	$(36 + 12)/2 = 24$	$(36 - 12)/2 = 12$
Wednesday	$(36 + 16)/2 = 26$	$(36 - 16)/2 = 10$
Thursday	$(50 - 10)/2 = 20$	$(50 + 10)/2 = 30$
Friday	$(40 - 8)/2 = 16$	$(40 + 8)/2 = 24$

Answer: B

Since, the total quantity of oranges sold on Wednesday = 26

So, the quantity of fresh oranges sold on Wednesday = $26 * (8/13) = 16$

And the quantity of stale oranges sold on Wednesday = $26 - 16 = 10$

Since, the total quantity of apples sold on Wednesday = 10

So, the quantity of fresh apples sold on Wednesday = $10 * (3/5) = 6$

And the quantity of stale apples sold on Wednesday = $10 - 6 = 4$

Since, the total quantity of fresh fruits sold on Wednesday = $16 + 6 = 22$

And the total quantity of stale fruits sold on Wednesday = $10 + 4 = 14$

So, the required percentage = $[(22 - 14)/14] * 100 = 57.14\%$ (approx.)

Hence, option B is correct.

17. Questions

Days	Total quantity of oranges and apples sold together (in kg)	The difference between Quantity of sold oranges and the Quantity of sold apples
Monday	$60 + 14 = 74$	$30 - 16 = 14$
Tuesday	$30 + 6 = 36$	$22 - 10 = 12$
Wednesday	$30 + 6 = 36$	$24 - 8 = 16$
Thursday	$40 + 10 = 50$	$30 - 20 = 10$
Friday	$32 + 8 = 40$	$24 - 16 = 8$

Days	Quantity of sold oranges (in kg)	Quantity of sold apples (in kg)
Monday	$(74 - 14)/2 = 30$	$(74 + 14)/2 = 44$
Tuesday	$(36 + 12)/2 = 24$	$(36 - 12)/2 = 12$
Wednesday	$(36 + 16)/2 = 26$	$(36 - 16)/2 = 10$
Thursday	$(50 - 10)/2 = 20$	$(50 + 10)/2 = 30$
Friday	$(40 - 8)/2 = 16$	$(40 + 8)/2 = 24$

Answer: E

From I:

The amount received by selling apples on Tuesday = $12 * 50 = \text{Rs.}600$

The amount received by selling apples on Thursday = $30 * 40 = \text{Rs.}1200$

Since, the required ratio = $600:1200 = 1:2$

So, I is true.

From II:

The amount received by selling oranges on Tuesday = $24 * 45 = \text{Rs.}1080$

The amount received by selling oranges on Thursday = $20 * 50 = \text{Rs.}1000$

Since, the required total = $(1080 + 1000) = \text{Rs.}2080$

So, II is true.

From III:

The amount received by selling oranges on Tuesday = $24 * 45 = \text{Rs.}1080$

The amount received by selling apples on Thursday = $30 * 40 = \text{Rs.}1200$

Since, the required percentage = $(1080/1200) * 100 = 90\%$

So, III is true.

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Hence, all are true.

Hence, option E is correct.

18. Questions

Days	Total quantity of oranges and apples sold together (in kg)	The difference between Quantity of sold oranges and the Quantity of sold apples
Monday	$60 + 14 = 74$	$30 - 16 = 14$
Tuesday	$30 + 6 = 36$	$22 - 10 = 12$
Wednesday	$30 + 6 = 36$	$24 - 8 = 16$
Thursday	$40 + 10 = 50$	$30 - 20 = 10$
Friday	$32 + 8 = 40$	$24 - 16 = 8$

Days	Quantity of sold oranges (in kg)	Quantity of sold apples (in kg)
Monday	$(74 - 14)/2 = 30$	$(74 + 14)/2 = 44$
Tuesday	$(36 + 12)/2 = 24$	$(36 - 12)/2 = 12$
Wednesday	$(36 + 16)/2 = 26$	$(36 - 16)/2 = 10$
Thursday	$(50 - 10)/2 = 20$	$(50 + 10)/2 = 30$
Friday	$(40 - 8)/2 = 16$	$(40 + 8)/2 = 24$

Answer: D

Since, the quantity of apples sold on Friday = 24 kg

And the quantity of oranges sold on Friday = 16 kg

So, $M = [(24 - 16)/16] * 100 = 50\%$

$M = 50$

Since, the quantity of apples sold on Wednesday = 10 kg

And the quantity of oranges sold on Wednesday = 26 kg

So, $N = [(26 - 10)/10] * 100 = 160\%$

$N = 160$

From I:

$M = (N/2) - 20$

$50 = (160/2) - 20$

$50 \neq 60$

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So, relation given in I is not correct.

From II:

$$3M - N = 10$$

$$3 * 50 - 160 = 10$$

$$-10 \neq 10$$

So, relation given in II is not correct.

From III:

$$N/10 = (M/10) + 10$$

$$160/10 = (50/10) + 10$$

$$16 \neq 15$$

So, relation given in III is not correct.

Hence, none is correct.

Hence, option D is correct.

19. Questions

Days	Total quantity of oranges and apples sold together (in kg)	The difference between Quantity of sold oranges and the Quantity of sold apples
Monday	$60 + 14 = 74$	$30 - 16 = 14$
Tuesday	$30 + 6 = 36$	$22 - 10 = 12$
Wednesday	$30 + 6 = 36$	$24 - 8 = 16$
Thursday	$40 + 10 = 50$	$30 - 20 = 10$
Friday	$32 + 8 = 40$	$24 - 16 = 8$

Days	Quantity of sold oranges (in kg)	Quantity of sold apples (in kg)
Monday	$(74 - 14)/2 = 30$	$(74 + 14)/2 = 44$
Tuesday	$(36 + 12)/2 = 24$	$(36 - 12)/2 = 12$
Wednesday	$(36 + 16)/2 = 26$	$(36 - 16)/2 = 10$
Thursday	$(50 - 10)/2 = 20$	$(50 + 10)/2 = 30$
Friday	$(40 - 8)/2 = 16$	$(40 + 8)/2 = 24$

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Answer: E

Since, the quantity of oranges sold on Monday = 30 kg

And the quantity of apples sold on Monday = 44 kg

So,

$$\Rightarrow [30 * X]/[44 * (X + Y)] = 6/11$$

$$\Rightarrow 5X = 4X + 4Y$$

$$\Rightarrow X = 4Y$$

$$\Rightarrow X/Y = 4/1$$

$$\Rightarrow X: Y = 4: 1$$

Hence, option E is correct.

20. Questions

Days	Total quantity of oranges and apples sold together (in kg)	The difference between Quantity of sold oranges and the Quantity of sold apples
Monday	$60 + 14 = 74$	$30 - 16 = 14$
Tuesday	$30 + 6 = 36$	$22 - 10 = 12$
Wednesday	$30 + 6 = 36$	$24 - 8 = 16$
Thursday	$40 + 10 = 50$	$30 - 20 = 10$
Friday	$32 + 8 = 40$	$24 - 16 = 8$

Days	Quantity of sold oranges (in kg)	Quantity of sold apples (in kg)
Monday	$(74 - 14)/2 = 30$	$(74 + 14)/2 = 44$
Tuesday	$(36 + 12)/2 = 24$	$(36 - 12)/2 = 12$
Wednesday	$(36 + 16)/2 = 26$	$(36 - 16)/2 = 10$
Thursday	$(50 - 10)/2 = 20$	$(50 + 10)/2 = 30$
Friday	$(40 - 8)/2 = 16$	$(40 + 8)/2 = 24$

Answer: C

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When the quantities of apples sold on each of the given days are described in a pie chart:

The sum of the pie chart = $44 + 12 + 10 + 30 + 24 = 120$ kg

Since, the degree distribution of the quantity of apples sold on Monday = $360 * (44/120) = 132^\circ$

And the degree distribution of the quantity of apples sold on Friday = $360 * (24/120) = 72^\circ$

So, the required difference = $132^\circ - 72^\circ = 60^\circ$

Hence, option C is correct.